

Recurrent Events

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Introduction

Recurrent-event analysis applies whenever a single subject can experience the event of interest multiple times during follow-up — hospital admissions, seizures, infections, asthma exacerbations, recurrent tumours. Treating only the first event collapses information and wastes power; treating all events as independent inflates effective sample size and produces invalid standard errors. Several modelling frameworks balance these extremes, each suited to a different scientific question.

Prerequisites

A working understanding of Cox regression, counting-process data formats, and within-subject correlation as a source of dependent observations.

Theory

The main approaches:

- **Andersen-Gill (AG)**: a counting-process Cox model that treats all events equally and uses cluster-robust standard errors to account for within-subject dependence. Estimates a single baseline hazard for the recurrent process.
- **Prentice-Williams-Peterson (PWP)**: stratifies by event number, allowing separate baseline hazards for the first, second, third, etc. event. Two flavours: total time (gap from study start) and gap time (gap from previous event). Use when the hazard structure changes with event order.
- **Wei-Lin-Weissfeld (WLW)**: fits a separate marginal Cox model for each event order, using only the first event for $K = 1$, the first two for $K = 2$, etc. Conceptually cleaner but increasingly out of favour because it conditions on having survived enough events to be at risk.
- **Shared-frailty**: a Cox model with a subject-level random effect u_i that captures unmeasured individual heterogeneity. Combine with AG when frailty is the natural interpretation of within-subject correlation.

Assumptions

The chosen correlation structure matches the data, the hazard specification (PH within strata or marginally) is appropriate, and counting-process intervals are correctly constructed (no overlaps, no gaps).

R Implementation

```
library(survival)

# Counting-process data: id, tstart, tstop, status (1 = event)
data(cgd, package = "survival")

# Andersen-Gill
ag <- coxph(Surv(tstart, tstop, status) ~ treat + inherit,
            data = cgd, cluster = id)
summary(ag)
```

```
# PWP stratified by event number
cgd$event.num <- ave(cgd$status, cgd$id, FUN = cumsum) + 1
pwp <- coxph(Surv(tstart, tstop, status) ~ treat + inherit + strata(event.num),
             data = cgd, cluster = id)
summary(pwp)
```

Output & Results

Both fits return hazard-ratio estimates with cluster-robust standard errors. AG produces a single HR per covariate covering the full recurrent-event process; PWP produces strata-specific baselines while sharing covariate effects across strata, allowing different baseline hazards per event order.

Interpretation

A reporting sentence: “An Andersen-Gill counting-process Cox model estimated that prophylactic treatment reduced the recurrent-infection hazard by 60 % (HR 0.40, 95 % CI 0.25 to 0.63), with cluster-robust SEs accounting for within-subject correlation; the PWP model stratified by event order produced a consistent estimate (HR 0.43, 95 % CI 0.26 to 0.70), suggesting the treatment effect does not vary materially across event order.” Compare AG and PWP whenever the data structure permits.

Practical Tips

- AG is the default starting point for most applications and the easiest to interpret.
- PWP (gap-time) is appropriate when the hazard depends on time since the previous event; PWP (total time) when it depends on time since study start.
- WLW has fallen out of favour because it conditions on event order in a way that produces awkward causal interpretations.
- For unmeasured individual heterogeneity not captured by covariates, add a shared frailty term to the AG model (+ `frailty(id)`).
- Always use cluster-robust SEs (`cluster = id`); ignoring within-subject correlation produces SEs that are too small and Type I error inflated.
- Pair hazard ratios with absolute counts of events, expected rates, and cumulative mean functions; recurrent-event audiences typically reason in absolute frequencies rather than hazard ratios.

R Packages Used

`survival` for `coxph()` with `cluster()` and `strata()`; `frailtypack` for joint frailty models with terminal events and recurrent events fitted simultaneously; `reReg` for additional recurrent-event methods including marginal mean models; `mets` for clustered survival data with flexible dependence structures.

For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren’t.
- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

See also — labs in this chapter

- Survival primer: KM, log-rank, Cox PH
- Time-varying covariates and immortal-time bias
- Competing risks and multistate models

- Survival ML (random survival forest, DeepSurv)
- Time-dependent Brier, IPA, external validation